Computer Practical

Parcel Modelling

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1 Overview

In this computer practical, you will use a parcel model to investigate how water vapour condenses onto aerosol particles to form cloud droplets. As in previous practicals, the model advances a set of coupled ordinary differential equations in time.

By the end of the practical you should be able to:

- explain why a rising air parcel cools and eventually forms cloud;
- describe how vertical velocity and initial humidity affect cloud formation;
- explain how aerosol number, size-distribution width and particle size affect cloud droplet number;
- configure, run and analyse a compiled environmental model.

2 Model formulation

This model calculates the properties of an air parcel as it moves vertically through the atmosphere. Atmospheric pressure changes with height; expansion associated with decreasing pressure cools a rising parcel and can eventually lead to condensation. The model solves several coupled equations:

$$e_0 = 610.7 \exp\left(\frac{L_v}{R_v}\left[\frac{1}{273.15} - \frac{1}{T}\right]\right), \tag{1}$$

$$e_v = e_0 \exp\left(\frac{2\sigma_w}{R_v T \rho_l r_i}\right), \tag{2}$$

$$\frac{dP}{dt} = -\rho_a g w, \tag{3}$$

$$\frac{dT}{dt} = \frac{R_a}{c_p}\frac{T}{P}\frac{dP}{dt} + \frac{L_v}{c_p}\frac{dw_l}{dt}, \tag{4}$$

$$\frac{dm_i}{dt} = 4\pi r_i D_v \left(w_{v,\infty}\rho_a - \frac{e_v(T)}{R_v T}\right), \tag{5}$$

$$\frac{dw_l}{dt} = \sum_i^N N_i \frac{dm_i}{dt}, \tag{6}$$

$$\frac{dw_{v,\infty}}{dt} = -\frac{dw_l}{dt}. \tag{7}$$

Here e_0 and e_v are the equilibrium vapour pressures over a flat and curved surface respectively; L_v is the latent heat of vaporisation; R_v and R_a are the specific gas constants for water vapour and air; T is temperature; σ_w is the surface tension of water; ρ_l and ρ_a are the densities of liquid water and air; r_i is the radius of an aerosol particle or droplet; P is pressure; g is gravity; c_p is the specific heat capacity of air; m_i is the mass of a single aerosol particle or droplet; $w_{v,\infty}$ is the water-vapour mixing ratio; N_i is the number of aerosol particles or droplets; w is the vertical wind; and D_v is a diffusivity.

These are the equations that the Fortran model will solve for us.

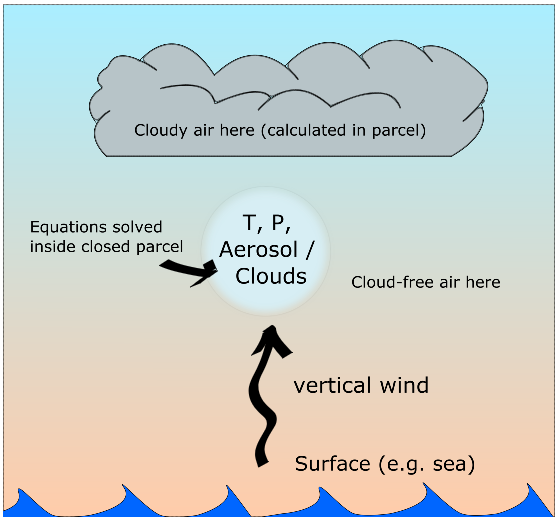

Figure 1: Schematic of the parcel model. The equations are solved for the properties of the rising parcel; assuming the surrounding air behaves similarly allows the output to be interpreted along a vertical coordinate.

3 Aerosol size distributions

Atmospheric aerosol particle sizes are often represented using log-normal distributions. Three parameters describe a log-normal mode: number concentration N, median diameter $\bar{D}$ and geometric standard deviation σ_g. Atmospheric aerosol populations commonly contain several modes, such as nucleation, accumulation and coarse modes, so the input file contains three modes for each aerosol type.

In the Marine Cloud Brightening geoengineering concept, adding sea-salt aerosol to the background aerosol may alter cloud droplet number and cloud reflectivity. This is the idea we will explore in the practical.

The parcel model is written in Fortran and follows an adiabatic parcel along a vertical coordinate. It generates output in NetCDF format, a common data format in atmospheric and environmental sciences. A Python script reads this output and produces the plots used in the exercises. Figure 1 summarises the model setup.

4 Downloading and compiling the code

Log into the server using an SSH window and, in a separate window, log in using SFTP. Arrange the windows so that you can move easily between them.

If you have not already downloaded the practical code, type:

```
git clone https://github.com/EnvModelling/bin-microphysics-model
```

This downloads the repository into your current working directory.

Move into the directory (you can type `cd bin-` followed by the Tab key to auto-complete), then inspect its contents and compile the Fortran code:

```
cd bin-microphysics-model
ls
make
```

The `make` command compiles the Fortran source into machine code that the computer can run.

5 Viewing and editing the input file

The model input file is called `namelist.in`. Open it in `nano` with line numbers displayed:

```
nano -l namelist.in
```

Save and exit using `Ctrl-X`, then `Y`, followed by `Enter`.

Rather than relying only on line numbers, you can search within `nano` using `Ctrl-W`. For example, search for `winit`, `rhinit` or `n_aer1`. In the current version, `winit` is on line 18, `rhinit` is on line 28, and the aerosol size-distribution parameters are on lines 161–168.

The arrays `n_aer1`, `d_aer1` and `sig_aer1` contain 3×2 values: three log-normal submodes for each of two externally mixed aerosol types. The first three values describe the background aerosol; the second three describe sea-salt aerosol.

6 Running the model

After the code has been compiled (Section 4), the normal workflow is:

1. edit `namelist.in` to configure the model;
2. run the model using `./run.sh`.

If you change any Fortran source code, run `make` again before running the model.

Table 1: Parameters used in this practical. The aerosol arrays list the three background modes followed by the three sea-salt modes.

Variable	`Default value`	Description
`winit`	`0.3`	parcel vertical velocity (m s^{-1})
`tinit`	`290`	initial temperature (K)
`pinit`	`100000`	initial pressure (Pa)
`rhinit`	`0.65`	initial relative humidity (fraction)
`n_aer1`	`46.64e6, 153.42e6, 166.77e6` `0e6, 0e6, 0.e6`	number concentration in each mode (kg^{-1})
`d_aer1`	`18e-9, 39e-9, 154.e-9` `1e-6, 1e-6, 100.e-9`	number-median dry diameter in each mode (m)
`sig_aer1`	`0.348, 0.354, 0.465` `0.3, 0.3, 0.5`	$\ln \sigma_g$ for each mode (distribution width)

7 Experiments

Keep a practical record. Save each figure and make brief notes as you go. Record the parameter values used for each simulation and what changed in the output. You may be asked to interpret these experiments in the assessment.

7.1 Temperature lapse rate of the parcel

The first simulation is the default case: background aerosol, a parcel starting at the ground and rising at 0.3 m s^{-1} to a height of 1200 m. Run it with:

```
./run.sh
```

The model will create an output file. Check that it is present using:

```
ls /tmp/$USER/
```

Here `$USER` is expanded automatically by the Linux shell to your username.

Exercise. To plot the results, run:

```
python3 python/example_plot_bmm2.py
```

Python will create `vertical.png` in your temporary output directory. In the SFTP window, replace `YOUR_USERNAME` with your username and type:

```
get /tmp/YOUR_USERNAME/vertical.png
```

The figure shows the properties of the parcel along the vertical coordinate and will be discussed in the practical.

Exercise. From the top-right plot, estimate how rapidly temperature decreases with height up to about 800 m altitude. This lapse rate is characteristic of dry air rising approximately adiabatically.

Exercise. What happens to the rate of cooling above 800 m? Why?

Make a note of the liquid-water mixing ratio at cloud top and the approximate height at which cloud droplets first form.

7.2 Effect of vertical wind speed

Vertical velocity controls how rapidly the parcel cools with time and how quickly supersaturation develops, which can strongly affect droplet activation. Change `winit` (line 18 in the current `namelist.in`).

Run simulations at 10× less and 10× more than the default vertical velocity: `winit=0.03` and `winit=3.0`. Open the file using:

```
nano -l namelist.in
```

After each change, run:

```
./run.sh
```

and follow the plotting and download procedure in Section 7.1.

Exercise. How does the lapse rate between 0 and 800 m compare between these simulations?

Exercise. Look at both the liquid-water mixing ratio (top left) and CDNC (Cloud Drop Number Concentration). What effect does changing vertical velocity have? Why?

Exercise. Approximately what height does cloud start to form in these simulations? Why?

7.3 Increasing humidity (water vapour)

Return to the committed default state of the model:

```
git reset --hard
```

Find `rhinit` (line 28 in the current `namelist.in`) and change it to `rhinit=0.95`. Run the model, generate the plot and download it in the usual way.

Exercise. How did increasing the initial humidity change the result? Why?

7.4 Adding sea-salt aerosol

From the main `bin-microphysics-model` directory, return to the default:

```
git reset --hard
```

We will now investigate three properties of an added sea-salt mode: number concentration, distribution width and particle size. The simulations are cumulative: **do not reset the model between runs 1–3**.

Run	Sea-salt number	Median diameter	$\ln\sigma_g$
Default	0	100 nm	0.5
1	1000×10^6 kg^{-1}	100 nm	0.5
2	1000×10^6 kg^{-1}	100 nm	0.01
3	1000×10^6 kg^{-1}	800 nm	0.01

1. **Add sea-salt aerosol.** Change the third sea-salt number concentration in `n_aer1` (line 162) from zero to 1000×10^6 kg^{-1}:

   ```
   0e6, 0e6, 1000.e6,
   ```

   Keep the default third-mode diameter (`100.e-9`) and width (`0.5`). Run the simulation, make the plot and save the image.
2. **Narrow the sea-salt distribution.** Keep the sea-salt number concentration at `1000.e6` and the diameter at `100.e-9`. Change only the third sea-salt value of `sig_aer1` (line 168) from `0.5` to `0.01`:

```
0.3, 0.3, 0.01,
```

Run the simulation, make the plot and save the image.

3. **Increase the sea-salt particle size.** Keep the sea-salt number concentration at `1000.e6` and `sig_aer1` at `0.01`. Change only the third sea-salt value of `d_aer1` (line 165) from `100.e-9` to `800.e-9`:

```
1e-6, 1e-6, 800.e-9,
```

Run the simulation, make the plot and save the image.

Exercise. Compare run 1 with the default simulation in Section 7.1. What is the effect of adding 1000×10^6 kg^{-1} sea-salt particles?

Exercise. Compare run 2 with run 1. What is the effect of narrowing the sea-salt size distribution while keeping its number concentration and median diameter fixed?

Exercise. Compare run 3 with run 2. What is the effect of increasing the sea-salt median diameter from 100 nm to 800 nm while keeping number concentration and distribution width fixed?

7.5 Bonus: Ice formation

From the main `bin-microphysics-model` directory, return to the default with `git reset -hard`. Find `z_ctop` (line 98 in the current `namelist.in`) and change the cloud-top height to 12000 m (`z_ctop=12000,`). Run the simulation and make the plot. We will discuss the output in the practical.